PANAV GOHIL

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LinkedIn | GitHub

Education

Delhi Technological University | New Delhi, India 2025 – Present

B.Tech in Electronics Engineering (VLSI Design and Technology)

- Current CGPA: 7.5
- Academic Achievement: Secured 98.54 Percentile in JEE Main

Green Valley International Public School | New Delhi, India 2023 – 2025

Senior Secondary (CBSE, PCM) | Class XII: 92%

Presidium School & GD Goenka Public School | Haryana, India Till 2023

Class X (CBSE): 93.6%

Technical Skills

Languages: Python, C, C++, HTML, CSS, JavaScript

AI & ML: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, CNNs, Feature Engineering, Model Evaluation

Computer Vision: OpenCV, MediaPipe, FaceMesh, Image Processing, Real-Time Computer Vision, Gesture Recognition

Embedded Systems & Hardware: Arduino, ESP32-CAM, Sensor Integration, IoT Prototyping, Hardware Interfacing

Developer Tools/Platform: Git, GitHub, VS Code, Google Colab

Core Concepts: Data Structures & Algorithms, OOP, Problem Solving, Basic ML Pipelines

Professional Skills: Technical Research, Documentation, Public Speaking, Team Collaboration

Projects

DriveGuard AI – Intelligent Driver Monitoring & Safety System June 2026 – Present

– Python, TensorFlow, OpenCV, MediaPipe FaceMesh, CNN

- Developed an AI-powered driver monitoring system capable of detecting fatigue, distraction, and unsafe driving behavior in real time using Computer Vision and Deep Learning.
- Built and trained a Convolutional Neural Network (CNN) on a custom eye-state dataset to classify eye conditions and detect prolonged eye closure for fatigue assessment.
- Designed a multi-factor Driver Risk Score by combining eye closure duration, blink frequency, yawning patterns, and head orientation to improve fatigue prediction reliability.
- Implemented real-time audio alerts, visual warnings, and session analytics for fatigue event logging, attention tracking, and post-drive safety evaluation.

SENTRY: Reconnaissance & Hazard Detection Platform April 2026 – May 2026

– Arduino, ESP32-CAM, MQ2 Gas Sensor, Flame Sensor, PIR Sensor, Ultrasonic Sensor, Servo Motor

- Engineered an autonomous robotics system designed for real-time reconnaissance and environmental hazard detection.
- Integrated ultrasonic and PIR sensors for navigation and motion detection, alongside MQ2 gas and flame sensors for physical and chemical threat identification.
- Programmed a servo-mounted laser system and automated buzzer alerts to provide immediate feedback upon hazard detection, bridging hardware logic with practical safety applications.

HandPilot – Touchless Human-Computer Interaction System May 2026 – June 2026

– Python, OpenCV, MediaPipe, Scikit-learn, Random Forest, PyAutoGUI

- Developed a real-time hand gesture recognition and computer control system using OpenCV and MediaPipe to detect and track 21 hand landmarks from live webcam input.
- Built a custom dataset and engineered 42 relative-coordinate features for robust gesture classification independent of hand position.
- Trained and evaluated a Random Forest classifier achieving 98.75% accuracy across gestures including Open Palm, Fist, Point, and Thumbs Up.
- Integrated gesture-based cursor navigation, clicking, dragging, scrolling, volume, and brightness control to enable touchless human-computer interaction.

Achievements

- Secured 98.53 Percentile in JEE (Main), achieving an All India Rank (AIR) of 22,540 among 1.5+ million applicants.
- Grand Finalist (Top 40 Teams Nationally) at Hackfluence 2026 for developing 'EventLens'.
- Awarded 1st Position in DTU ALTAIR Ideathon for proposing and presenting an innovative technology-based solution.
- Represented DTU at prestigious national debating competitions, including IIT Bombay, IIT Delhi, and IGDTUW Parliamentary Debates.